20 SELECTIVE BREEDING

Ever since humans became farmers they have been selectively breeding animals and plants. This chapter looks at traditional methods of selective breeding and modern developments involving cloned organisms.

LEARNING OBJECTIVES

- Understand how selective breeding can be used to produce plants with desired characteristics
- Understand how selective breeding can be used to produce animals with desirable characteristics

BIOLOGY ONLY

- Describe the process of micropropagation (tissue culture) in which explants are grown *in vitro*

BIOLOGY ONLY

- Understand how micropropagation can be used to produce commercial quantities of genetically identical plants with desirable characteristics
- Describe the stages in the production of cloned mammals involving the introduction of a diploid nucleus from a mature cell into an enucleated egg cell, illustrated by Dolly the sheep
- Understand how cloned transgenic animals can be used to produce human proteins

About 12 000 years ago, the human way of life changed significantly. Humans began to grow plants and keep animals for milk and meat. They became farmers rather than hunters. This change first took place in the Middle East. Similar changes took place a little later in the Americas (where potatoes and maize were being grown) and in the Far East (where rice was first cultivated).

In the Middle East, humans first grew the cereal plants wheat and barley, and kept sheep and goats. Later, their livestock included cattle and pigs. Cultivating crops and keeping stock animals made it possible for permanent settlements to appear – life in villages began. Because of the more certain food supply, there was spare time, for the first time ever, for some people to do things other than hunt for food.

Ever since the cultivation of the first wheat and barley and the domestication of the first stock animals, humans have tried to obtain bigger yields from them. They cross-bred different maize plants (and barley plants) to obtain strains that produced more grain. They bred sheep and goats to give more milk and meat – selective breeding had begun. Today, animals and plants are bred for more than just food. For example, animals are used to produce a range of medicines and for research into the action of drugs.

Selective breeding is best described as the breeding of only those individuals with desirable features. It is sometimes called 'artificial selection', as human choice, rather than environmental factors, is providing the selection pressure (compare this with natural selection, described in Chapter 19).

The methods used today for selective breeding are very different from those used only 50 years ago. Modern gene technology makes it possible to create a new strain of plant within weeks, rather than years.

TRADITIONAL SELECTIVE BREEDING

PLANTS

Traditionally, farmers have bred crop plants of all kinds to obtain increased yields. Probably the earliest example of selective breeding was the cross-breeding of strains of wild wheat. The aim was to produce wheat with a much increased yield of grain and with shorter, stronger stems (Figure 20.1). This wheat was used to make bread.



1 About 11 000 years ago, two strains of wild wheat were cultivated by farmers. Initially, all attempts at cross-breeding to produce wheats with a better yield gave only sterile offspring.

2 About 8000 years ago, a fertile hybrid wheat appeared from these two wild wheats. This was called emmer wheat and had a much higher yield than either of the original wheats.

3 The emmer wheat was cross-bred with another wild wheat to produce wheat very similar to the wheats used today to make bread. This new wheat had an even bigger yield and was much easier to 'process' to make flour.

▲ Figure 20.1 Modern wheat is the result of selective breeding by early farmers.

DID YOU KNOW?

The production of modern bread wheats by selective breeding is probably one of the earliest examples of producing genetically modified food. Each original wild wheat species had 14 chromosomes per cell. The wild emmer hybrid had 28 chromosomes per cell. Modern bread wheat has 42 chromosomes per cell. Selective breeding has modified the genetic make-up of wheat.

Other plants have been selectively bred for certain characteristics. *Brassica* is a genus of cabbage-like plants. One species of wild brassica (*Brassica olera*) was selectively bred to give several strains, each with specific features (see Figure 20.2). Some of the strains had large leaves, others had large flower heads, and others produced large buds.



▲ Figure 20.2 Selectively breeding the original wild brassica plants to enhance certain features has produced several familiar vegetables.

Selective breeding has produced many familiar vegetables. Besides the ones produced from *Brassica*, selective breeding of wild *Solanum* plants has produced the many strains of potatoes that are eaten today. Carrots and

parsnips are also the result of selective breeding programmes. Crop plants are bred to produce strains that:

- give higher yields
- are resistant to certain diseases (the diseases would reduce the yields)
- are resistant to certain insect pest damage (the damage would reduce the yield)
- are hardier (so that they survive in harsher climates or are productive for longer periods of the year)
- have a better balance of nutrients in the crop (for example, plants that contain more of the types of amino acids needed by humans).

DID YOU KNOW?

Plant breeders do not just breed plants for food. Nearly all garden flowers are the result of selective breeding. Breeders have selected flowers to have a particular size, shape, colour and fragrance. Roses and orchids are among the most selectively bred of our garden plants.

Figure 20.3 shows a field of potato plants. Some have been bred to be resistant to insect pests, while others were not selectively bred in this way.



▲ Figure 20.3 Selective breeding can reduce damage by pests. The plants on the right have been bred to be resistant to a fungal pest. Plants on the left are not resistant to the pest.

ANIMALS

Farmers have bred stock animals for similar reasons to the breeding of crops. They have selected for animals that:

- produce more meat, milk or eggs
- produce more fur or better quality fur
- produce more offspring
- show increased resistance to diseases and parasites.

Again, like crop breeding, breeding animals for increased productivity has been practised for thousands of years. A stone tablet found in Iran appears to record the results of breeding domesticated donkeys. It was dated at over 5000 years old.

For many thousands of years, the only way to improve livestock was to mate a male and a female with the features that were desired in the offspring. In cattle, milk yield is an important factor and so high-yielding cows would be bred with bulls from other high-yielding cows.

Since about 1950, the technique of **artificial insemination (AI)** has become widely available. Bulls with many desirable features are kept and semen is obtained from them. The semen is diluted, frozen and stored. Farmers can buy quantities of this semen to inseminate their cows. the semen is transferred into the cow's uterus using a syringe. AI makes it possible for the semen from one prize bull to be used to fertilise many thousands of cows.

Modern sheep are domesticated wild sheep, and pigs have been derived from wild boars. Just think of all the varieties of dogs that now exist. All these have been derived from one ancestral type. This original 'dog' was a domesticated wolf (Figure 20.4). In domesticating the wolf, humans gained an animal that was capable of herding stock animals. The sheepdog has all the same instincts as the wolf except the instinct to kill. This has been selectively 'bred out'.



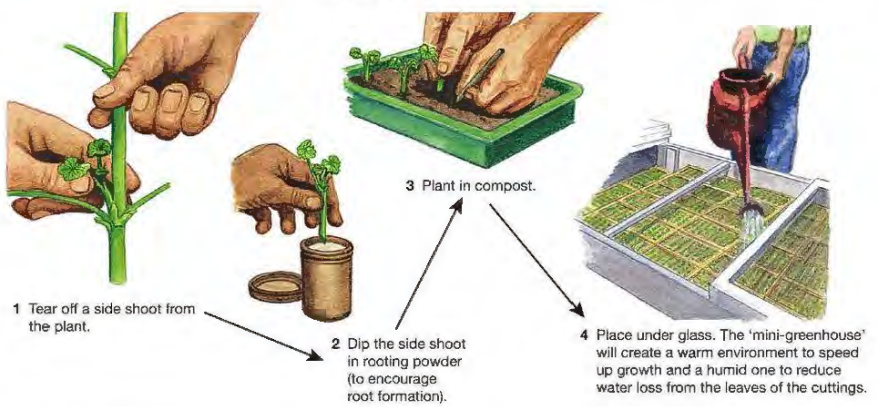
▲ Figure 20.4 The many different breeds of dog all originate from a common ancestor – the wolf.

BIOLOGY ONLY

MODERN SELECTIVE BREEDING

CLONING PLANTS

The term cloning describes any procedure that produces genetically identical offspring. Taking cuttings of plants and growing them is a traditional cloning technique (Figure 20.5).



▲ Figure 20.5 Taking stem cuttings

All the cuttings contain identical genes as they are all parts of the same parent plant. As they grow, they form new cells by mitosis, copying the genes in the existing cells exactly. The cuttings develop into a group of genetically identical plants – a **clone**. Any differences will be due to the environment. Many garden flowers have traditionally been propagated this way.

Some modern cloning techniques are essentially the same as taking cuttings – removing pieces of a plant and growing them into new individuals. The technology, however, is much more sophisticated. By using the technique of **micropropagation**, thousands of plants can quickly be produced from one original (Table 20.1).

Table 20.1 The main stages in micropropagation

Stages	Illustrations
The tips of the stems and side shoots are removed from the plant to be cloned. These parts are called **explants**. The explants are trimmed to a size of about 0.5–1 mm, and surface-sterilised to kill any microorganisms. They are then placed in a sterile agar medium that contains nutrients and plant hormones to encourage growth (Figure 20.6). More explants can be taken from the new shoots that form on the original ones. This can be repeated until there are enough to supply the demand.	 ▲ Figure 20.6 Explants growing in a culture medium.
The explants with shoots are transferred to another culture medium containing a different balance of plant hormones to induce root formation (Figure 20.7).	 ▲ Figure 20.7 Explants forming roots.
When the explants have grown roots, they are transferred to greenhouses and transplanted into compost (Figure 20.8). They are then gradually acclimatised to normal growing conditions. The atmosphere in the greenhouse is kept very moist to reduce water loss from the young plants. Because of the amount of water vapour in the air, they are often called 'fogging greenhouses'.	 ▲ Figure 20.8 Young plants being grown in compost in a greenhouse.

DID YOU KNOW?

Many strains of bananas are infertile. They are now commonly reproduced by micropropagation. Other plants produced this way include lilies, orchids and agave plants.

There are many advantages to propagating plants in this way:

- large numbers of genetically identical plants can be produced rapidly
- species that are difficult to grow from seed or from cuttings can be propagated by this method
- plants can be produced at any time of the year
- large numbers of plants can be stored easily (many can be kept in cold storage at the early stages of production and then developed as required)
- genetic modifications can be introduced into thousands of plants quickly, affer modifying only a few plants.

CLONING ANIMALS

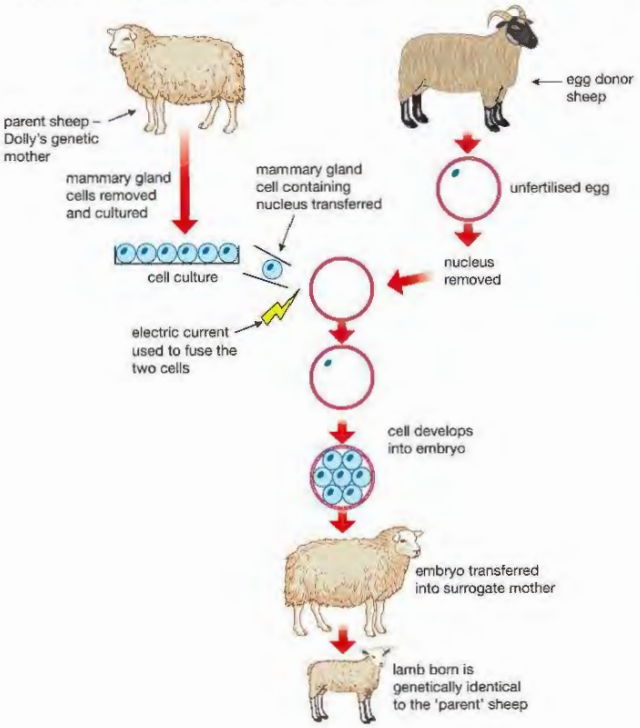
KEY POINT

The nucleus of an ovum is haploid (see Chapter 18). It cannot develop into a new individual because it only has half the chromosomes of normal body cells. An ordinary diploid body cell, even though it has all the chromosomes, is too specialised. Transferring a diploid nucleus into an egg cell that has had its nucleus removed creates a cell that is capable of developing into a new individual.

We have been able to clone plants by taking cuttings for thousands of years. It is now possible to make genetically identical copies of animals. The first, and best-known, example of this is the famous cloned sheep, Dolly.

Dolly was cloned from a body cell of an adult sheep (Figure 20.9). Scientists first took an ovum (egg cell) from a donor sheep and removed its nucleus, producing an *enucleated* cell. They then took cells from the mammary (milk-producing) gland of a second 'parent' sheep (Dolly's genetic mother) and cultured them in a special solution that kept them alive but stopped them growing. Next they placed one of the mammary gland cells next to the enucleated cell and fused the two cells together using an electric current. The nucleus from the mammary gland cell was now inside the enucleated cell.

They monitored the resulting cell for several days to watch its development. When it started to divide they transferred the embryo into the uterus of another sheep, a 'surrogate mother', to complete its development. After 148 days, Dolly was born. Dolly's cells contained exactly the same genetic information as those of the body cell from the 'parent' sheep.



▲ Figure 20.9 How 'Dolly' was produced.



▲ Figure 20.10 Inserting a mammary gland cell into an egg cell that has had its nucleus removed.

Dolly was only produced after many unsuccessful attempts. Since then, the procedure has been repeated using other sheep as well as rats, mice and pigs. Some of the animals produced are born deformed. Some do not survive to birth. Biologists believe that these problems occur because the genes that are transferred to the egg are 'old genes'. These genes came from an animal that had already lived for several years and from cells specialised to do things other than produce sex cells. It will take much more research to make the technique reliable.

DID YOU KNOW?

In 2016 (20 years after the birth of Dolly) the University of Nottingham in the UK announced that they had cloned several more sheep, including four called Debbie, Denise, Dianna and Daisy. These four animals were derived from the same cell line that produced Dolly, so were exact genetic copies of her. At the time of publication of the research, the sheep were nine years old and had developed no age-related health problems.

USING CLONED ANIMALS TO MAKE PROTEINS

Cloning animals has special value if the animal produces an important product. Sheep have been genetically modified to produce several human proteins (see Chapter 22). One of these is a protein called alpha-1-antitrypsin, which is used to treat conditions such as emphysema and cystic fibrosis. The genetically modified sheep secrete the protein in their milk. Cloning sheep like these would allow production of much more of this valuable protein. Polly, the first cloned, genetically modified sheep, was born a year after Dolly.

DID YOU KNOW?

Animals that have had genes transferred from other species are called *transgenic* animals.

END OF BIOLOGY ONLY

CHAPTER QUESTIONS

More questions on selective breeding can be found at the end of Unit 5 on page 277.

SKILLS CRITICAL THINKING (7)

1 Which of the following is another term for selective breeding?

A cloning

B natural selection

C artificial insemination

D artificial selection

BIOLOGY ONLY

SKILLS CRITICAL THINKING (6)

2 Which of the following best describes a clone?

A a transgenic organism

B an offspring where the genetic material in every cell is identical to that of both parents

C an offspring where the genetic material in every cell is identical to that of one parent

D a type of sheep

END OF BIOLOGY ONLY

(7)

3 Which of the following best describes artificial insemination?

A transplanting an embryo in the uterus

B taking sperm and placing it directly in the uterus

C selectively breeding healthy animals

D fertilisation of an egg in a test tube

BIOLOGY ONLY

(8)

4 Which of the following is the best description of how the first cloned mammal (Dolly the sheep) was produced?

A A mammary gland cell was fused with an enucleated egg, grown into an embryo and allowed to develop in a surrogate mother.

B A surrogate mother had an egg removed, which was fused with a mammary gland cell and allowed to develop into an embryo.

C An egg was placed in an enucleated mammary gland cell, grown into an embryo and allowed to develop in a surrogate mother.

D An egg was placed in the mammary gland of a donor sheep and allowed to develop into an embryo.

END OF BIOLOGY ONLY

5 a How is selective breeding similar to natural selection?

b How is selective breeding different from natural selection?

BIOLOGY ONLY

SKILLS REASONING (8)

6 Selective breeding of crop plants often aims to increase the yield of the crop.

a Describe, and explain the reasons for, three other aims of selective breeding programmes in crop plants.

(7)

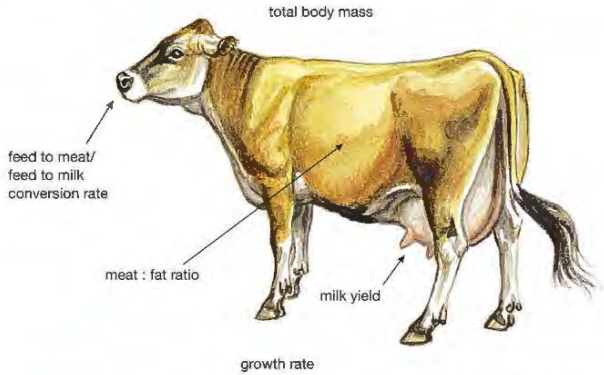
b Describe two advantages of micropropagation over the more traditional technique of taking cuttings.

c Explain why plants produced by micropropagation will be genetically identical to each other and to the parent plant.

END OF BIOLOGY ONLY

SKILLS ANALYSIS (7)

7 The diagram shows some of the features of a cow that might be used as a basis for a breeding programme.



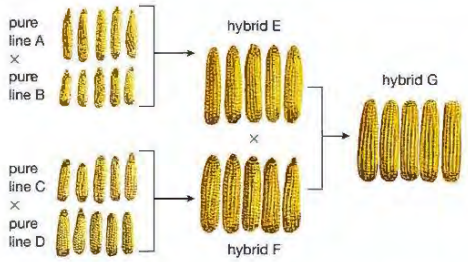
a Which features would you consider important in a breeding programme for dairy cattle?

SKILLS REASONING (8)

b Assume that you had all the techniques of modern selective breeding available to you. Describe how you would set about producing a herd of high-yielding beef cattle.

SKILLS ANALYSIS (9)

8 The diagram shows the results of a breeding programme to improve the yield of maize (sweetcorn).



a Describe the breeding procedure used to produce hybrid G.

(6) b Describe three differences between the corn cobs of hybrid G and those of C.

SKILLS REASONING (9) c How could you show that the differences between hybrid G and hybrid C are genetic?

SKILLS INTRAPERSONAL (9)

9 Carry out some research and write an essay about the benefits and concerns of selective breeding of animals. You should write about one side of A4.

UNIT QUESTIONS

SKILLS REASONING (9) **1**

For natural selection to operate, some factor has to exert a 'selection pressure'. In each of the following situations, identify both the selection pressure and the likely result of this selection pressure.

a Near old copper mines, the soil becomes polluted with copper ions that are toxic to most plants. **(2)**

b In the Serengeti of Africa, wildebeest are hunted by lions. **(2)**

c A farmer uses a pesticide to try to eliminate pests of a potato crop. **(2)**

(Total 6 marks)

BIOLOGY ONLY

SKILLS CRITICAL THINKING (7) **2**

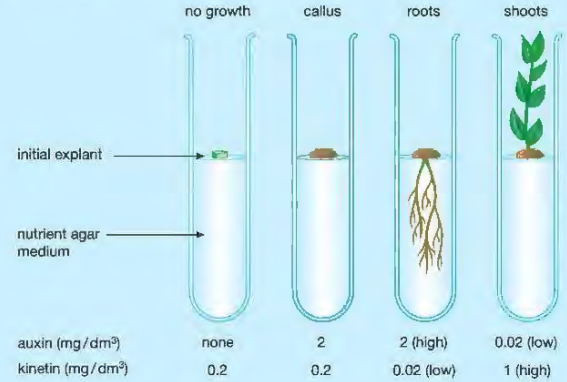
Micropropagation produces thousands of genetically identical plants. Small 'explants' from the parent plant are grown in culture media.

a Outline the main stages in micropropagation. **(4)**

SKILLS REASONING (7)

b Explain why the plants formed by micropropagation are genetically identical. **(2)**

c In some cases, the explants used contain only a few cells, without roots or shoots. Plant hormones are added to the culture media to encourage root and shoot formation. Two of these hormones are called kinetin and auxin. The diagram shows the effects of using different concentrations of the two hormones on root and shoot growth of the explants.



	no growth	callus	roots	shoots
auxin (mg/dm³)	none	2	2 (high)	0.02 (low)
kinetin (mg/dm³)	0.2	0.2	0.02 (low)	1 (high)

(7) i What is the effect of adding kinetin or auxin without any other hormone? **(2)**

ii Describe how you would treat these explants to produce first shoots and then roots. **(3)**

SKILLS CRITICAL THINKING (7) d Explain one advantage and one disadvantage of micropropagation. **(2)**

(Total 13 marks)

END OF BIOLOGY ONLY